

CIT338: Data Visualization

Week 1: Introduction to Data
Visualization



About Me



Ishani Udeshika

Experience:

Data Scientist at **Zone24x7 pvt Ltd**
Visiting Lecturer

Credentials:

- + M.Sc in Applied Statistics (*reading*)
- + B.Sc in Applied Mathematics and Computing
- + Microsoft Certified Azure Data Scientist





Course Overview



About You

<https://docs.google.com/forms/d/1XobpF9QipKNhHLNZLRFfExlyckfP9VABDWfb9zvpt5s/edit>

What is Data Visualization?

- Graphical representation of data
- Simplifies complex information
- Highlights patterns and outliers
- Enables effective communication



What are the DV elements?





M median

Qualitative



NOMINAL

Quantitative

Visualization



MEAN

Summary Statistics

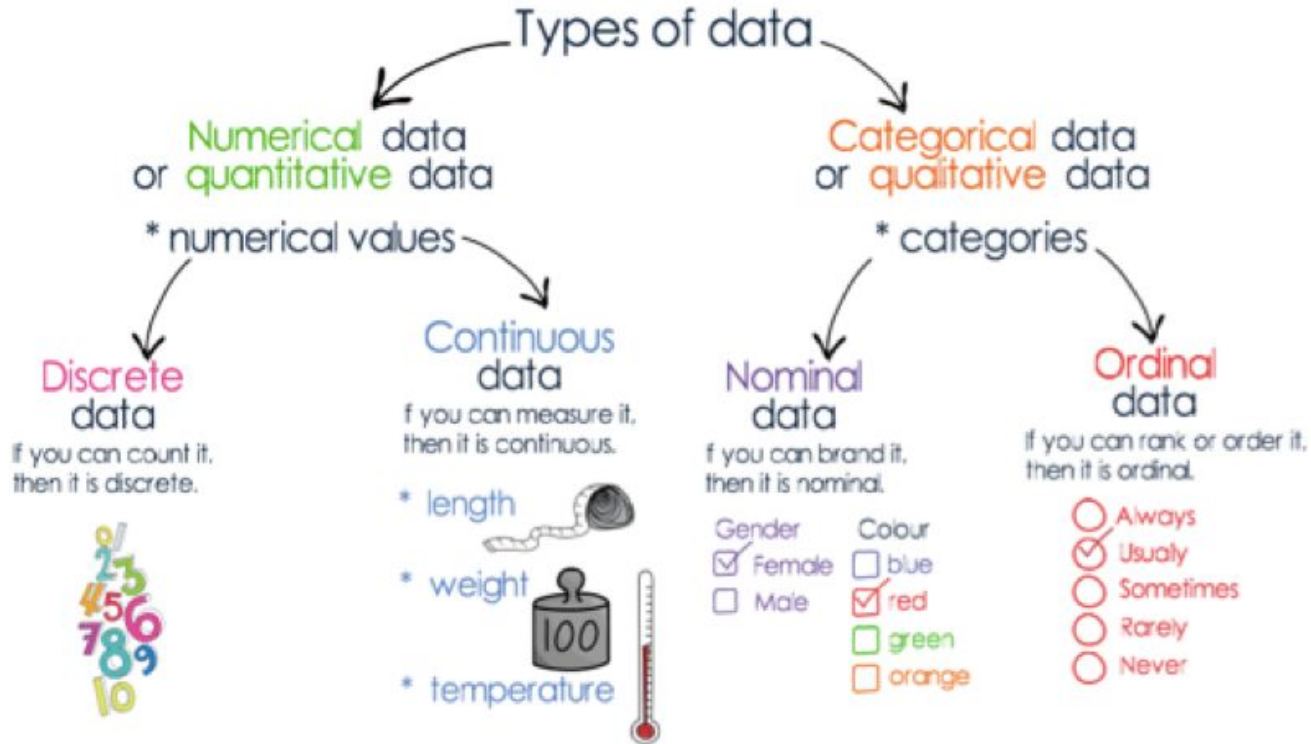


CATEGORIES

1 2 3 4



Understanding Data Types





Activity 1

My best friend has one older sibling
and two younger siblings

Quantitative

Qualitative

My best friend has curly brown hair

Age in months or years

Quantitative

Qualitative

The boys have brown, black, and red
hair

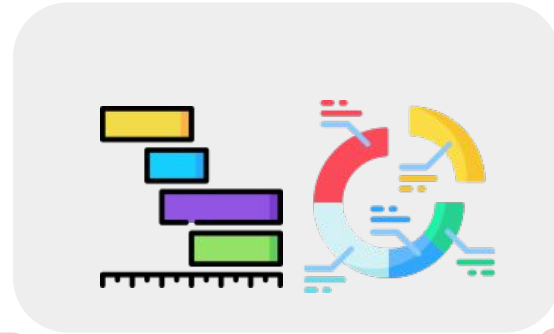
The cake was 50 percent chocolate
and 50 percent vanilla

Quantitative

Qualitative

Grades of Students in a Classroom

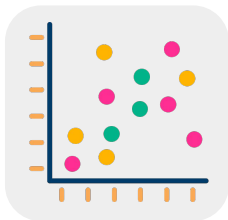
What Would You Like to Show?



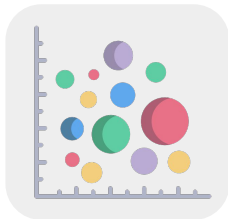
Relationship ?

Comparison ?

Distributions ?



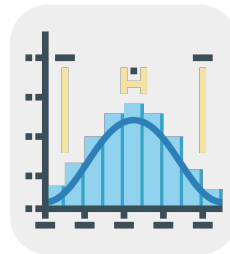
Scatter Plot



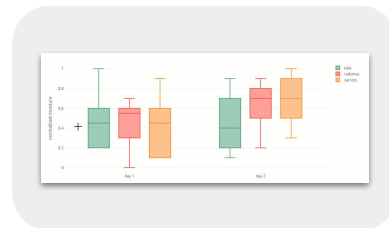
Bubble Chart



Histogram



Column Chart

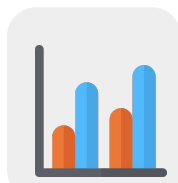


Box Plot

Among Items



Bar Chart



Column Chart



Pie Chart

Over Time



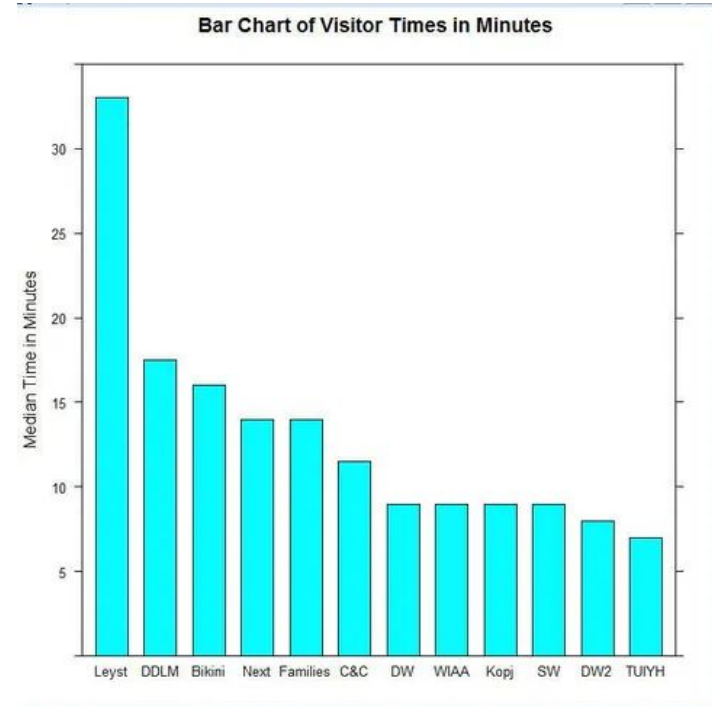
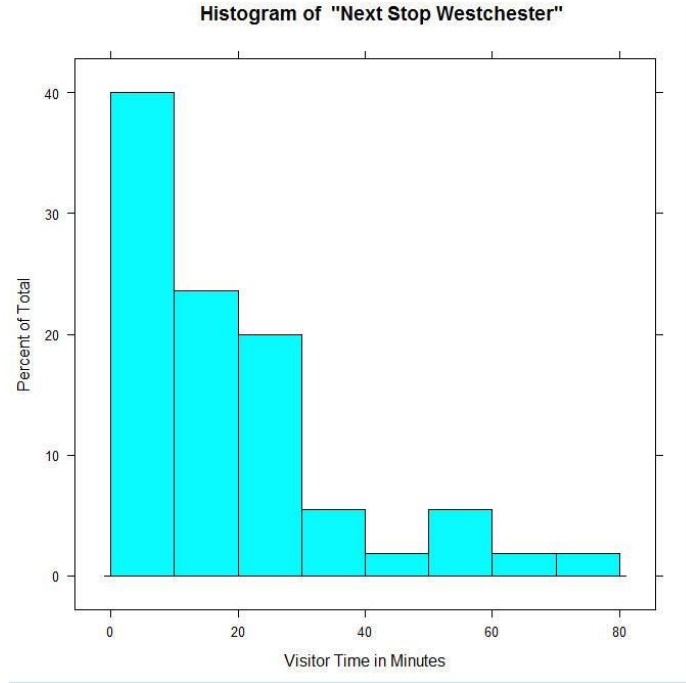
Single Line Chart



Multi Line Chart

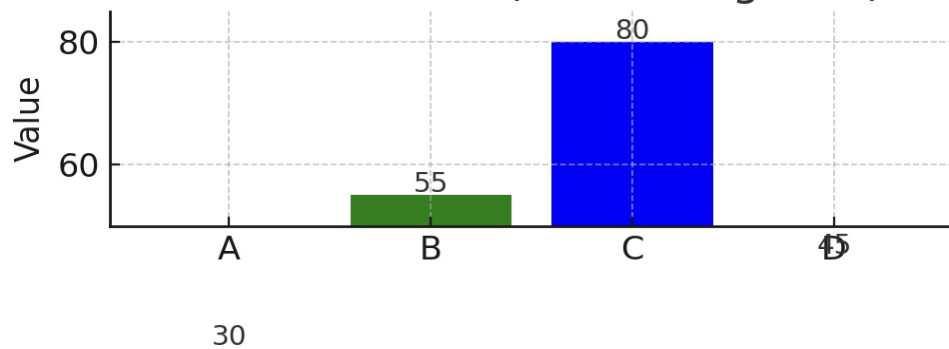
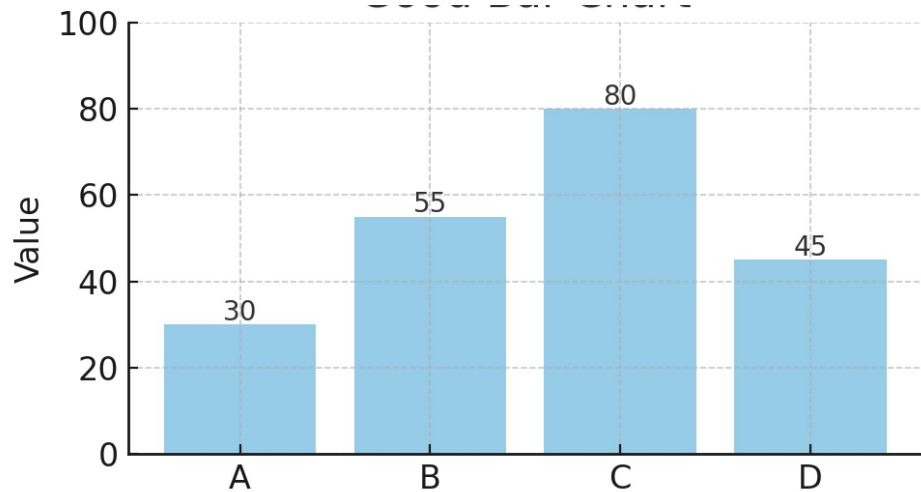
A Histogram is NOT a Bar Chart

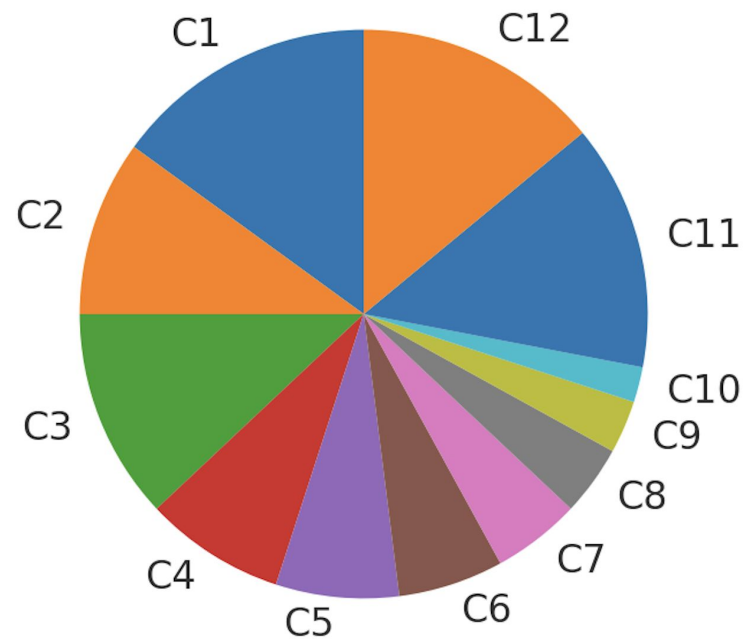
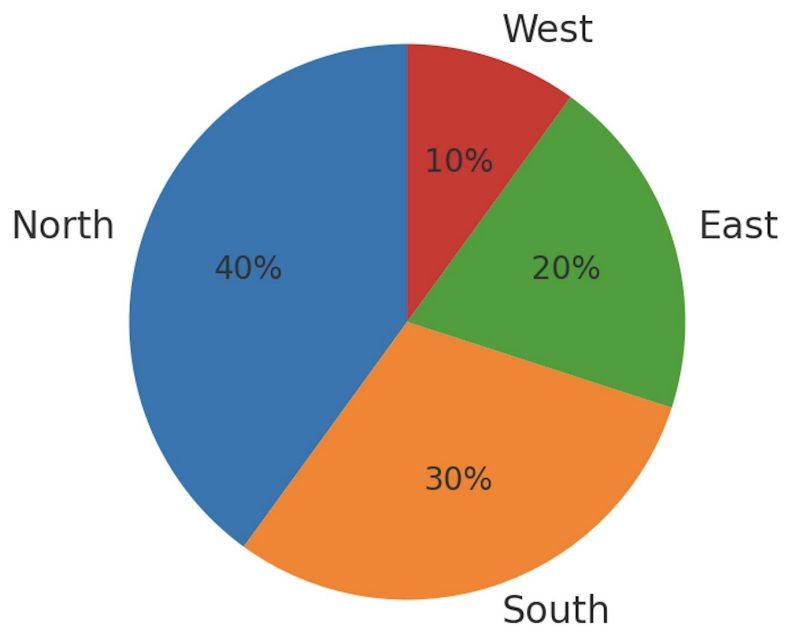
Do you agree with me?

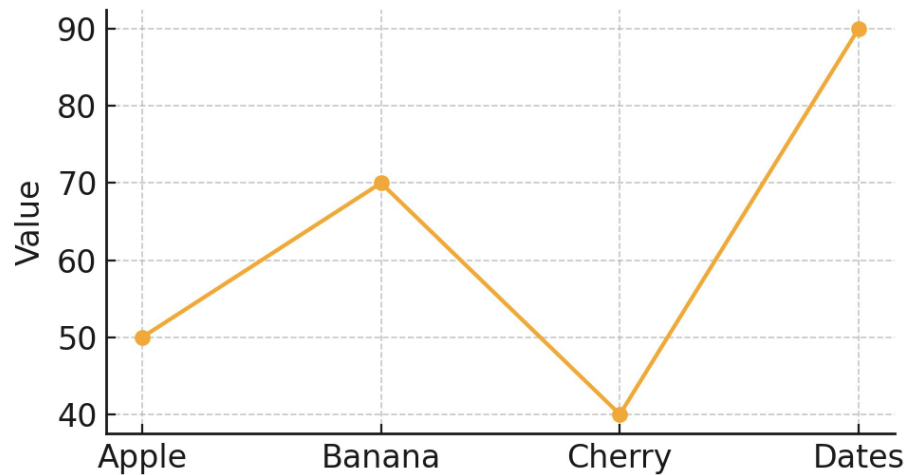
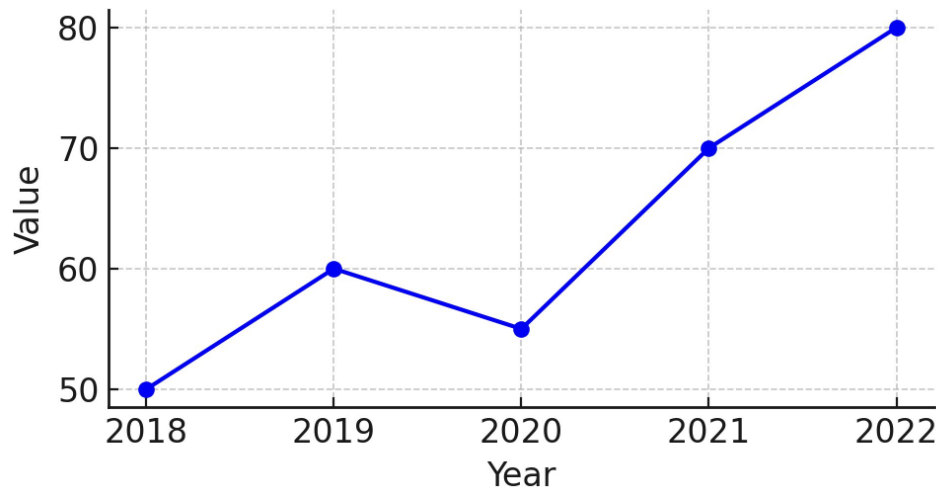




Bad vs Good Visualizations









Real-World Applications

- Healthcare: COVID-19 dashboards
- Business: Sales & marketing dashboards
- Sports: Player stats, heatmaps
- Public Policy: Budget allocation, crime stats



Quick Recap

https://docs.google.com/forms/d/e/1FAIpQLSc_eShxSdJSWw2xCr6Ayt27-sadAnw7js7vYMtCMY1p9qNmQw/viewform?usp=header



QnA



Thank you!